

CS410 Midterm 2 Part (a)

9 December 2022

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I. MULTIPLE CHOICE QUESTIONS (50 POINTS)

Given the following CFG C_1 . Answer questions 1-6 accordingly.

$$\begin{aligned} E &\rightarrow E + T \mid T \\ T &\rightarrow T * F \mid F \\ F &\rightarrow (E) \mid a \end{aligned}$$

1. What is V ?

- a) $\{E\}$
- b) $\{E, T\}$
- c) $\{E, T, F\}$
- d) $\{E, T, F, a\}$
- e) $\{E, T, F, a, *, +, (,)\}$

2. What is Σ ?

- a) $\{E\}$
- b) $\{E, T, F\}$
- c) $\{E, T, F, a, *, +, (,)\}$
- d) $\{a, *, +, (,)\}$
- e) $\{E, T, F, a\}$

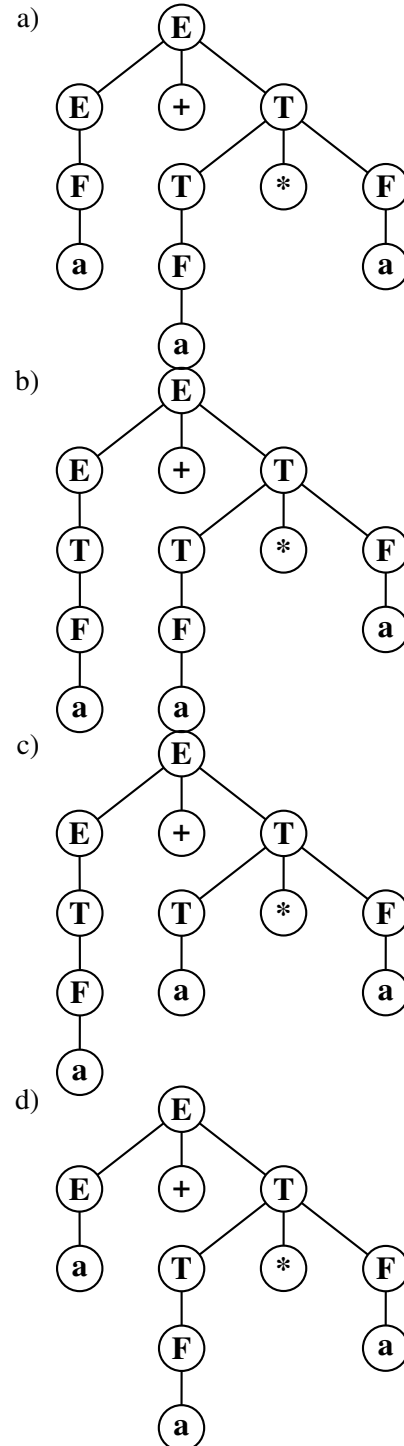
3. What is the start variable?

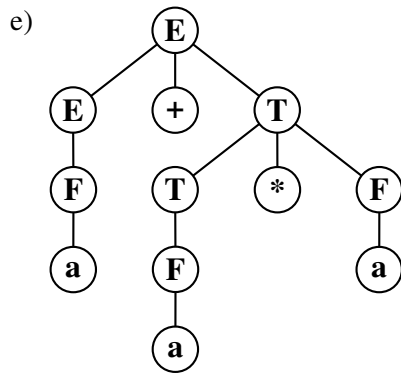
- a) E
- b) T
- c) F
- d) a
- e) *

4. Which of the following string is not accepted by C_1 ?

- a) a
- b) a * a
- c) a + a
- d) (a + a)
- e) All of them accepted

5. Which of the following is the parse tree for a + a * a?





6. Which of the following is the correct derivation for $a + a * a$?

- $E \rightarrow E + T \rightarrow F + T \rightarrow a + T \rightarrow a + T * F \rightarrow a + F * F \rightarrow a + a * F \rightarrow a + a * a$
- $E \rightarrow E + T \rightarrow T + T \rightarrow F + T \rightarrow a + T \rightarrow a + T * F \rightarrow a + F * F \rightarrow a + a * F \rightarrow a + a * a$
- $E \rightarrow E + T \rightarrow T + T \rightarrow F + T \rightarrow a + T \rightarrow a + T * F \rightarrow a + a * F \rightarrow a + a * a$
- $E \rightarrow E + T \rightarrow a + T \rightarrow a + T * F \rightarrow a + F * F \rightarrow a + a * F \rightarrow a + a * a$
- $E \rightarrow E + T \rightarrow F + T \rightarrow a + T \rightarrow a + T * F \rightarrow a + F * F \rightarrow a + a * F \rightarrow a + a * a$

7. Which of the following CFG generates the language $L = \{a^{2n}c^n \mid n \geq 0\}$?

- $S \rightarrow aaSc \mid \epsilon$
- $S \rightarrow aaSc \mid aac$
- $S \rightarrow aSc \mid \epsilon$
- $S \rightarrow aSc \mid ac$
- None

8. Which of the following CFG generates the language $L = \{a^n b^{n+2} \mid n \geq 1\}$?

- $S \rightarrow aSbbb \mid \epsilon$
- $S \rightarrow aSb \mid T$
- $S \rightarrow aSb \mid T$
 $T \rightarrow b$
- $S \rightarrow aSb \mid T$
 $T \rightarrow bb$
- None

9. Which of the following CFG generates the language $L = \{a^n b^m \mid n < m\}$?

- $S \rightarrow aSb \mid b$
- $S \rightarrow aSb \mid T$
 $T \rightarrow bb$
- $S \rightarrow aSb \mid T$
 $T \rightarrow bT \mid b$
- All of them correct
- None

10. What is the language generated by the following CFG?

$$S \rightarrow aSc \mid T \mid \epsilon$$

$$S \rightarrow aTb \mid \epsilon$$

- $L = \{a^n b^n c^n \mid n \geq 0\}$
- $L = \{a^n b^m c^n \mid n, m \geq 0\}$
- $L = \{a^n b^m c^k \mid n, m, k \geq 0\}$
- $L = \{a^n b^m c^k \mid n, m, k \geq 0, n = m + k\}$
- None

11. What is the language generated by the following CFG?

$$S \rightarrow aSbb \mid T$$

$$T \rightarrow Tb \mid \epsilon$$

- $L = \{a^n b^m \mid n, m \geq 0, m \geq 2n\}$
- $L = \{a^n b^m \mid n, m \geq 0, m \geq n\}$
- $L = \{a^n b^n \mid n \geq 0\}$
- $L = \{a^n b^m \mid n, m \geq 0\}$
- None

12. What is the language generated by the following CFG?

$$S \rightarrow 0S1 \mid A \mid B$$

$$A \rightarrow 0A \mid 0$$

$$B \rightarrow 1B \mid 1$$

- $L = \{0^n 1^n \mid n \geq 0\}$
- $L = \{0^m 1^n \mid n \geq 0, m \geq 0\}$
- $L = \{0^m 1^n \mid m \neq n\}$
- $L = \{0^m 1^n \mid n \leq m\}$
- None

13. Which of the following languages is ambiguous, that is it has at least two different leftmost derivations?

a)

$$E \rightarrow E + T \mid T$$

$$T \rightarrow T * F \mid F$$

$$F \rightarrow (E) \mid a$$

b)

$$E \rightarrow E + E \mid E * E \mid a$$

c)

$$S \rightarrow 0S \mid 1A$$

$$A \rightarrow 0A \mid \epsilon$$

- All of them are ambiguous
- None

14. Given the following ambiguous CFG, which of the following strings has two different leftmost derivations?

$$\begin{aligned} S &\rightarrow 0S \mid 1A \mid 0B0 \\ A &\rightarrow 0A \mid \epsilon \\ B &\rightarrow 0B0 \mid 1 \end{aligned}$$

- a) 1
- b) 010
- c) 11
- d) 01
- e) None

15. Given the following ambiguous CFG, which of the following strings has two different leftmost derivations?

$$\begin{aligned} S &\rightarrow AB \mid BA \\ A &\rightarrow A0 \mid \epsilon \\ B &\rightarrow 1B \mid \epsilon \end{aligned}$$

- a) 01
- b) 10
- c) 00
- d) All of them
- e) None

16. Which of the following CFG is in CNF?

a)

$$S \rightarrow aSb \mid b$$

b)

$$\begin{aligned} S &\rightarrow aSb \mid T \\ T &\rightarrow bb \end{aligned}$$

c)

$$\begin{aligned} S &\rightarrow aSb \mid T \\ T &\rightarrow bT \mid b \end{aligned}$$

- d) All of them in CNF
- e) None of them in CNF

17. If we have $k > 2$ variables on the right hand side of a rule R , how many rules should we add as extra instead of R if we want to convert R into CNF?

- a) $2k$
- b) $k + 2$
- c) $k - 1$
- d) $k + 1$
- e) $k + 3$

18. Which of the following CFG is in CNF?

a)

$$\begin{aligned} S &\rightarrow 0 \mid AB \mid CD \\ C &\rightarrow AB \\ D &\rightarrow AA \\ A &\rightarrow 0 \\ B &\rightarrow 1 \end{aligned}$$

b)

$$\begin{aligned} S &\rightarrow AC \mid CC \mid CE \\ E &\rightarrow DC \\ C &\rightarrow BB \\ D &\rightarrow BD \mid DA \mid a \mid b \\ A &\rightarrow a \\ B &\rightarrow b \end{aligned}$$

c)

$$\begin{aligned} S &\rightarrow AA \mid AB \mid 0 \mid 1 \mid \epsilon \\ A &\rightarrow 0 \\ B &\rightarrow 1 \end{aligned}$$

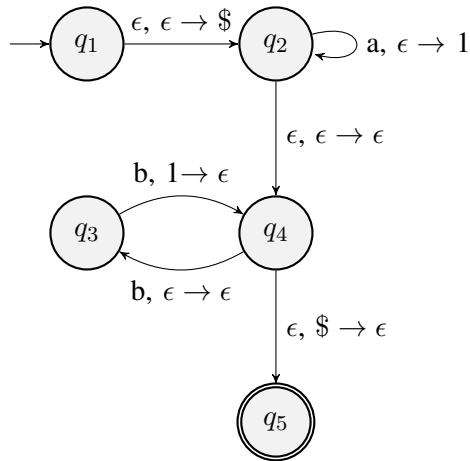
- d) All of them in CNF
- e) None of them in CNF

19. Given the following CFG, what is the resulting CFG after we apply ϵ removal rule in CNF conversion?

$$\begin{aligned} S &\rightarrow ASA \mid aB \\ A &\rightarrow B \mid S \\ B &\rightarrow b \mid \epsilon \end{aligned}$$

- a) $S \rightarrow a \mid aB \mid S$
 $A \rightarrow B \mid S$
 $B \rightarrow b$
- b) $S \rightarrow ASA \mid a \mid aB$
 $A \rightarrow B \mid S \mid \epsilon$
 $B \rightarrow b$
- c) $S \rightarrow ASA \mid a \mid aB \mid SA \mid AS \mid S$
 $A \rightarrow B \mid S$
 $B \rightarrow b$
- d) $S \rightarrow ASA \mid a \mid aB \mid S$
 $A \rightarrow B \mid S$
 $B \rightarrow b$
- e) None of them

Given the following PDA P_1 . Answer questions 20-21 accordingly.



20. How many rows and columns does the start transition table have?

- a) Rows: 5, Columns: 9
- b) Rows: 4, Columns: 9
- c) Rows: 5, Columns: 6
- d) Rows: 4, Columns: 6
- e) Rows: 5, Columns: 3

21. Which of the following is a string that is accepted by P_1 ?

- a) ab
- b) aabbbb
- c) aabb
- d) aaabbbbb
- e) aaabbbbbbb

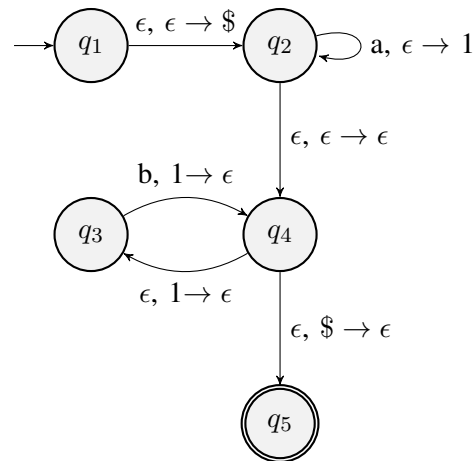
22. Which of the following languages is context-free?

- a) $L = \{a^n b^n \mid n \geq 0\}$
- b) $L = \{a^m b^n \mid n \geq 0, m \geq 0\}$
- c) $L = \{a^n b^{2n} \mid n \geq 0\}$
- d) $L = \{a^{2n} b^n \mid n \geq 0\}$
- e) All of them

23. Which of the following languages is not context-free?

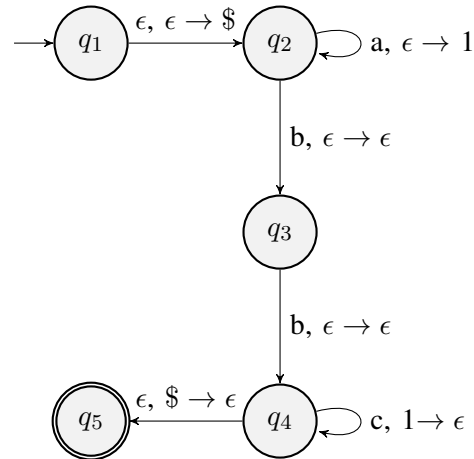
- a) $L = \{a^{2n} b^{n+1} b^{n-1} \mid n \geq 1\}$
- b) $L = \{a^n b^n c^{2n} d^{2n} \mid n \geq 0\}$
- c) $L = \{a^n b^{2m} c^{2n} d^{2m} \mid n, m \geq 0\}$
- d) $L = \{a^n b^{n^2} \mid n \geq 0\}$
- e) All of them not context-free

24. What is the language generated by the following PDA?



- a) $L = \{a^n b^n \mid n \geq 0\}$
- b) $L = \{a^m b^n \mid n \geq 0, m \geq 0\}$
- c) $L = \{a^n b^{2n} \mid n \geq 0\}$
- d) $L = \{a^{2n} b^n \mid n \geq 0\}$
- e) None

25. What is the language generated by the following PDA?



- a) $L = \{a^n b^m c^n \mid n, m \geq 0\}$
- b) $L = \{a^n b^2 c^n \mid n \geq 0\}$
- c) $L = \{a^n b^2 c^{2n} \mid n \geq 0\}$
- d) $L = \{a^{2n} b^2 c^n \mid n \geq 0\}$
- e) $L = \{a^n b^{2n} c^n \mid n \geq 0\}$